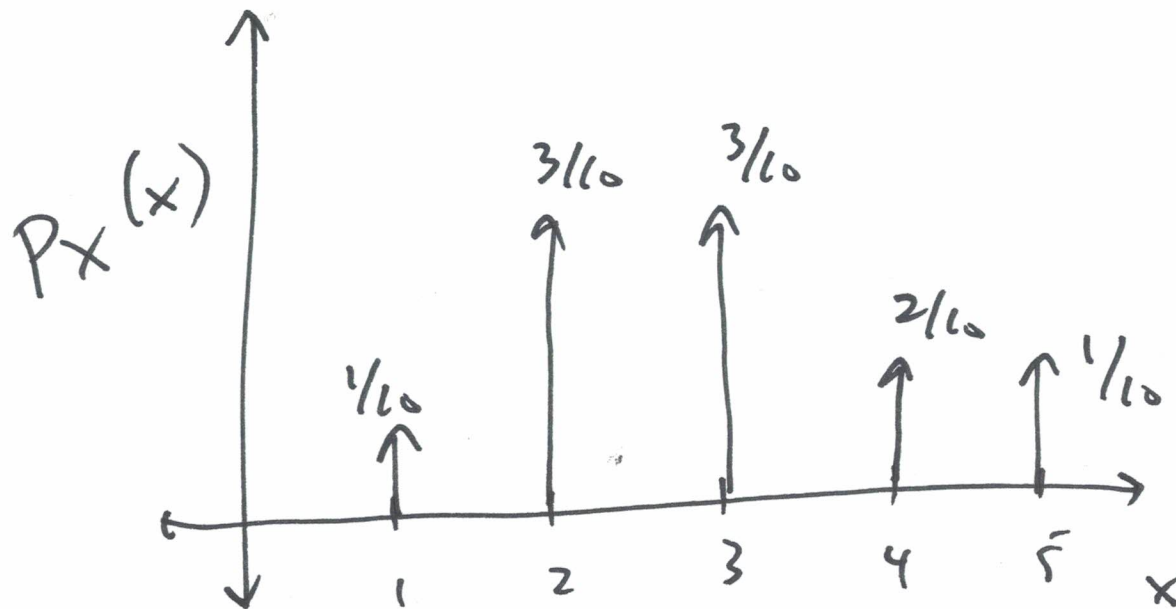


EXPECTED VALUE

LAST TIME: PMF (PROBABILITY MASS FUNCTION)

FOR A RV (RANDOM VARIABLE) X



THE EXPECTED VALUE

$$E(X) = \sum_{x \in S} x \cdot p_X(x)$$

↑
SUM OVER
ALL OUTCOMES
IN SAMPLE
SPACE

↑
VALUE OF
OUTCOME

↑
PROB. OF
GETTING
THAT
OUTCOME

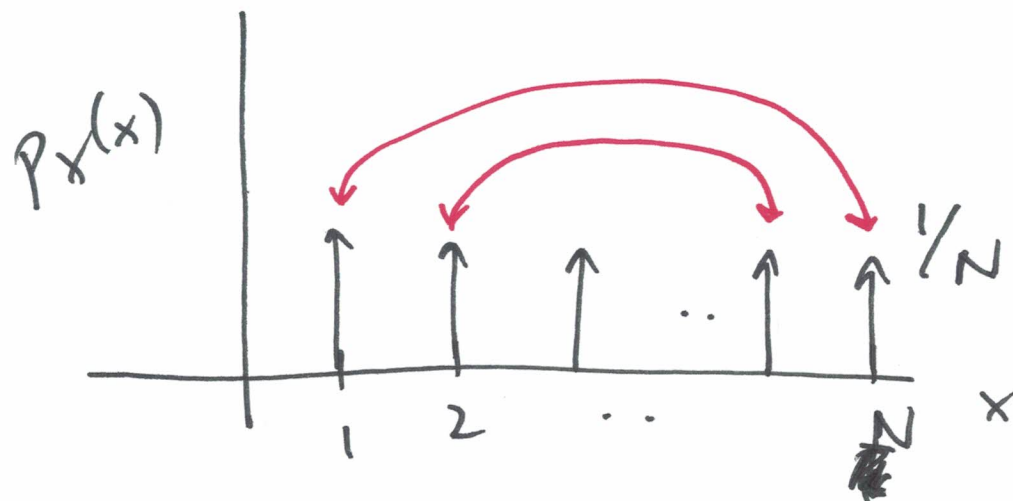
Ex:

$$\begin{aligned} & 1 \cdot \frac{1}{10} + 2 \cdot \frac{3}{10} + 3 \cdot \frac{3}{10} \\ & + 4 \cdot \frac{2}{10} + 5 \cdot \frac{1}{10} = \frac{1 + 6 + 9 + 8 + 5}{10} \\ & = \frac{29}{10} = 2.9 \end{aligned}$$

INTUITION: IF WE DID THE EXPERIMENT
MILLIONS OF TIMES, WHAT WOULD BE
THE AVERAGE OUTCOME?

NAMED PMFS:

UNIFORM ON $\{1, 2, \dots, N\}$



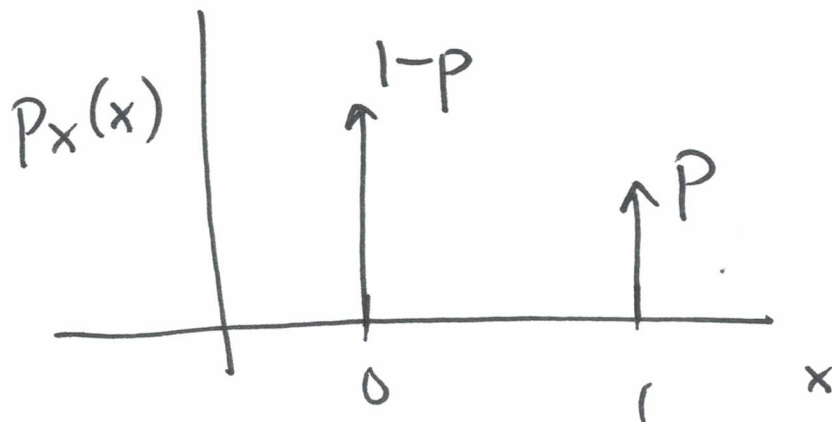
$$E(X) = \frac{1}{N} (1 + 2 + \dots + N)$$

$$\frac{1}{N} \frac{N}{2} (N+1)$$

$$= \frac{1}{2} (N+1)$$

BERNOULLI

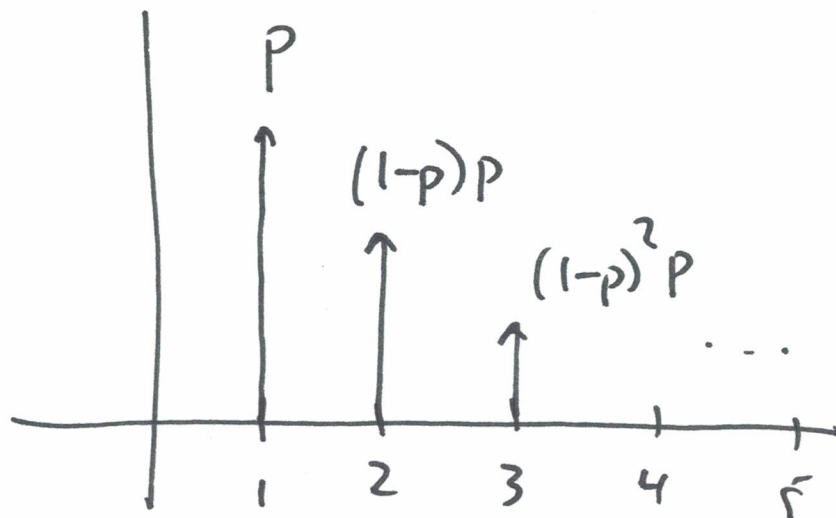
PMF:



$$E(x) = 0(1-p) + 1(p) = p.$$

GEOMETRIC

PMF:



$$p_X(x) = (1-p)^{k-1} p \quad k = \{1, 2, \dots, \infty\}$$

$$E(X) = \sum_{k=1}^{\infty} k (1-p)^{k-1} p$$

$$1 + a + a^2 + \dots = \frac{1}{1-a}$$

$$= p + 2(1-p)p + 3(1-p)^2 p + \dots$$

$$p(1 + (1-p) + (1-p)^2 + \dots) = p \cdot \frac{1}{1-(1-p)} = p \cdot \frac{1}{p} = 1$$

$$= p + (1-p)p + (1-p)^2 p + \dots$$

$$p(1-p)(1 + (1-p) + (1-p)^2 + \dots) = (1-p)p + (1-p)^2 p + \dots$$

$$((1-p)^2 p)(1 + \dots)$$

$$\left. \begin{array}{l} \frac{1}{p} \\ + \\ \frac{1-p}{1} \\ + \\ \frac{(1-p)^2}{(1-p)^2} \end{array} \right\}$$

$$= (p + (1-p)p + (1-p)^2 p + \dots)(1 +$$

$$p(1 + (1-p) + (1-p)^2 + \dots)(1 + (1-p) + (1-p)^2 + \dots)$$

$$= p \frac{1}{1-(1-p)} \cdot \frac{1}{1-(1-p)} = p \cdot \frac{1}{p} \cdot \frac{1}{p} = \frac{1}{p}.$$

$$E(X) = \frac{1}{p}.$$

IF p SMALL, $E(X)$ IS LARGE

BINOMIAL RANDOM VARIABLE:

$$P_X(k) = \binom{n}{k} (1-p)^{n-k} p^k \quad k=0, \dots, n$$

DIRECT COMPUTATION OF $E(X)$ NOT EASY.

USE PROPERTIES OF $E(X)$ TO MAKE
IT EASIER.

$$E(X + Y) = E(X) + E(Y)$$

$$E(aX) = a E(X)$$



BINOMIAL RV = SUM OF N BERNOLLI RVs.

$$\cancel{Y} = X_1 + \dots + X_n$$

BINOMIAL



BERNOLLI

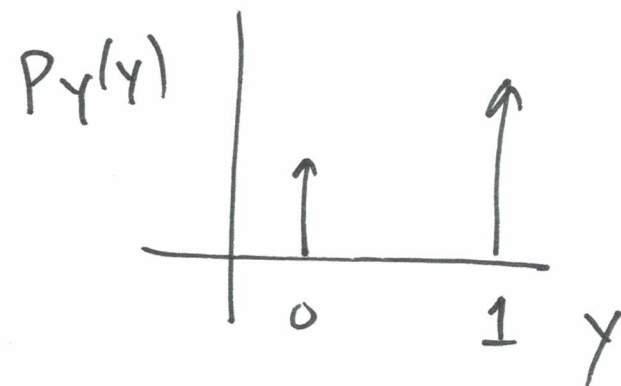
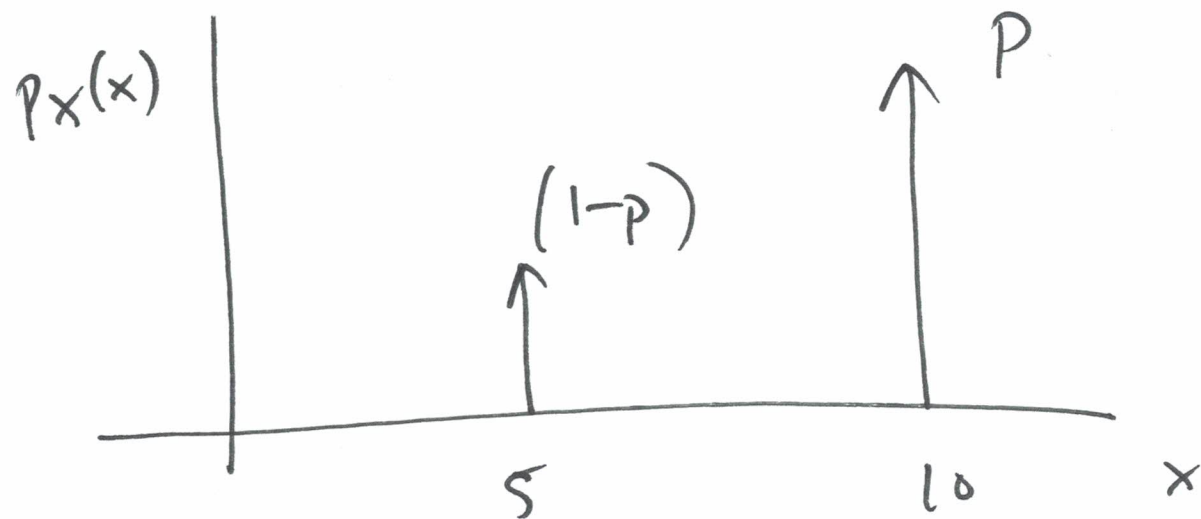
(SINGLE TRIALS)

$$E(Y) = E(X_1 + \dots + X_n)$$

$$= E(X_1) + \dots + E(X_n)$$

$$= p + \dots + p$$

$$= \boxed{np}$$



RELATED TO BERNOULLI RV Y BY

$$X = 5 + 5Y$$

$$E(X) = E(5 + 5Y)$$

$$= E(5) + E(5Y) = 5 + 5E(Y)$$

$$= 5(1-p) + 10p = 5 + 5p$$

$$= 5 + 5p$$

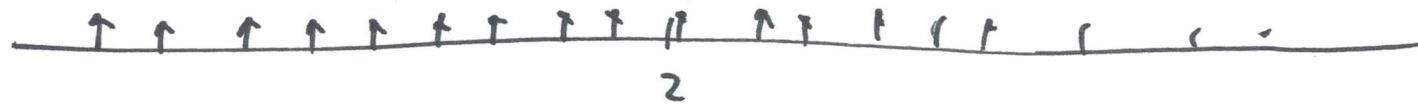
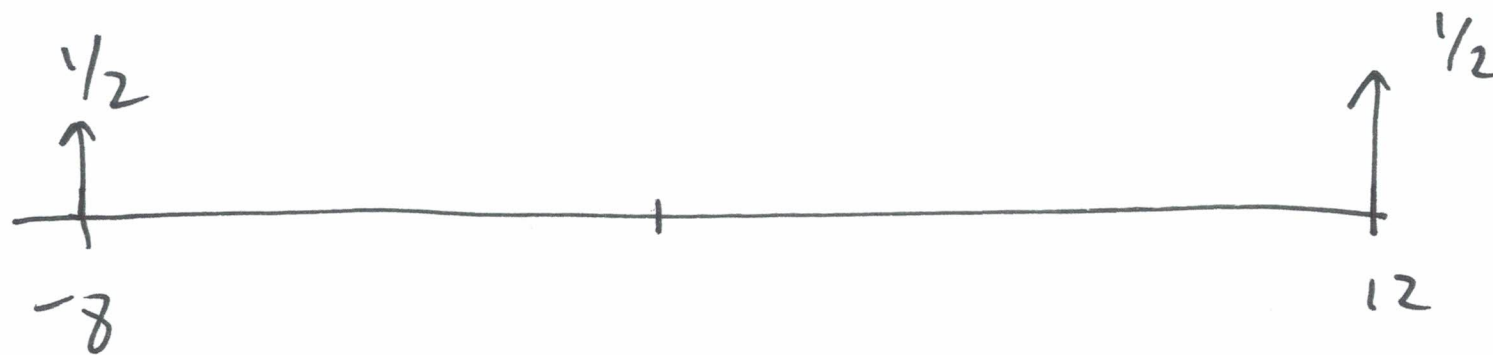
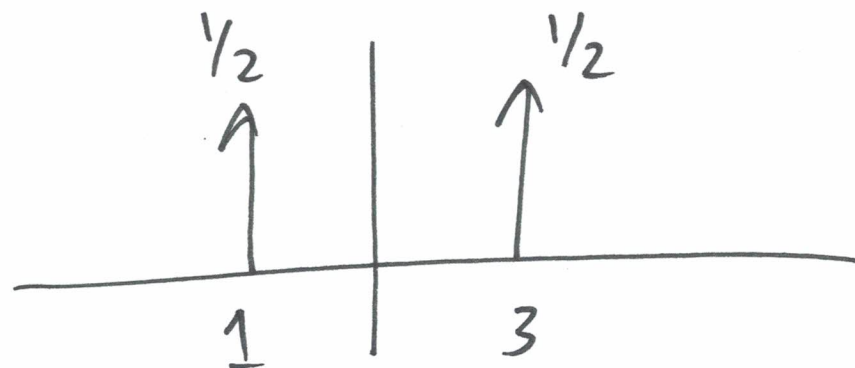
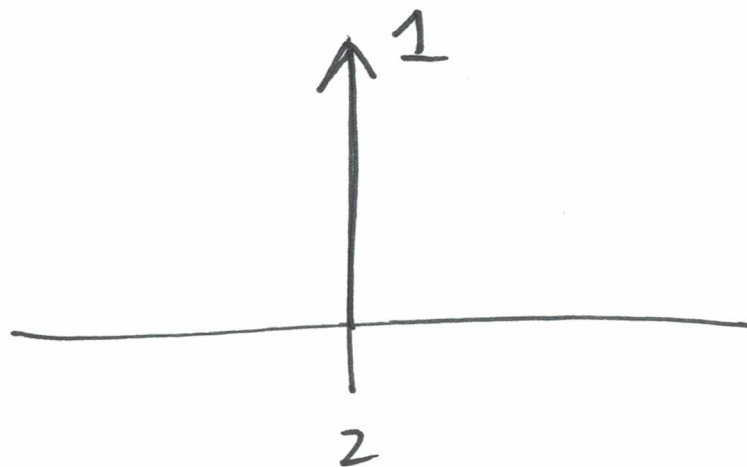
SAY IN FIRST EXAMPLE

X MODELS REWARD FOR PLAYING
A GAME.

How MUCH SHOULD YOU BE WILLING
TO ~~PAY~~ PAY TO PLAY?

$$E(X) = 2.9.$$

DON'T PAY MORE THAN \$2.90
TO PLAY, OTHERWISE YOU HAVE
AN EXPECTED LOSS!



VARIANCE: HOW FAR DO WE EXPECT
A RV TO BE FROM ITS MEAN?

$$\begin{aligned}\text{Var}(X) &= E\left((X - \underbrace{E(X)}_{\text{A NUMBER}})^2\right) \\ &= E\left((X - \mu_x)^2\right) \quad \text{A.K.A. } E(X)\end{aligned}$$

$$\begin{aligned}&= E\left(X^2 - 2\mu_x X + \mu_x^2\right) \\ E(X^2) - 2\mu_x^2 + \mu_x^2 &= E(X^2) - \underbrace{2\mu_x}_{-2\mu_x^2} \underbrace{E(X)}_{+ \mu_x^2} + E(\mu_x^2) \\ &= E(X^2) - \mu_x^2 = E(X^2) - (E(X))^2\end{aligned}$$

REMEMBER: $E(X^2) \neq (E(X))^2$

COMPUTING VARIANCE FOR EXAMPLE:

$$\begin{aligned} E(\underline{X^2}) &= \sum_x x^2 p_X(x) \\ &= \frac{1}{10}(1)^2 + \frac{3}{10}(2^2) + \frac{3}{10}(3^2) + \frac{2}{10}(4^2) + \frac{1}{10}(5^2) \\ &= \frac{1 + 12 + 27 + 32 + 25}{10} = \frac{97}{10} = 9.7 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= 9.7 - (2.9)^2 = 1.29. \end{aligned}$$

ALTERNATELY WE COULD COMPUTE

$$E((X - \mu_x)^2)$$

$$\begin{aligned} &= \frac{1}{10} ((1 - 2.9)^2) &= \frac{1}{10} (-1.9)^2 \\ &+ \frac{3}{10} ((2 - 2.9)^2) &+ \frac{3}{10} (-0.9)^2 \\ &+ \frac{3}{10} ((3 - 2.9)^2) &+ \frac{3}{10} (0.1)^2 \\ &+ \dots &+ \dots \end{aligned}$$

WILL GET SAME VALUE.

VARIANCE IS ALWAYS POSITIVE!

VARIANCES OF COMMON RANDOM VARIABLES
ARE LESS EASY TO COMPUTE BY HAND.

USE TABLES / PROPERTIES!

$$Y = aX$$

$$\text{Var}(aX) = \text{Var}(aX)$$

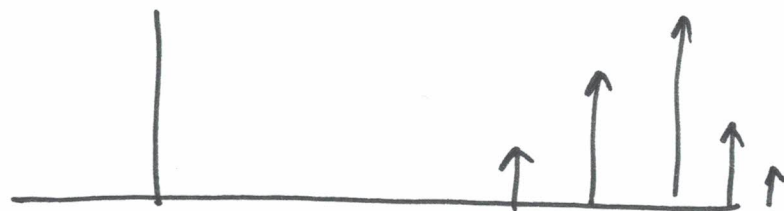
$$E(Y) = E(Y^2) - (E(Y))^2$$

$$= E((aX)^2) - (E(aX))^2$$

$$= a^2 E(X^2) - a^2 (E(X))^2$$

$$= a^2 \text{Var}(X)$$

$$\text{Var}(X + c) = \text{Var}(X)$$



$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

IF X AND Y ARE INDEPENDENT!

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$$

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$$

$$\text{Var}(X) = \sigma_x^2$$

$\sigma_x = \sqrt{\text{Var}(X)}$ IS THE
STANDARD DEVIATION.

MORE GENERALLY,

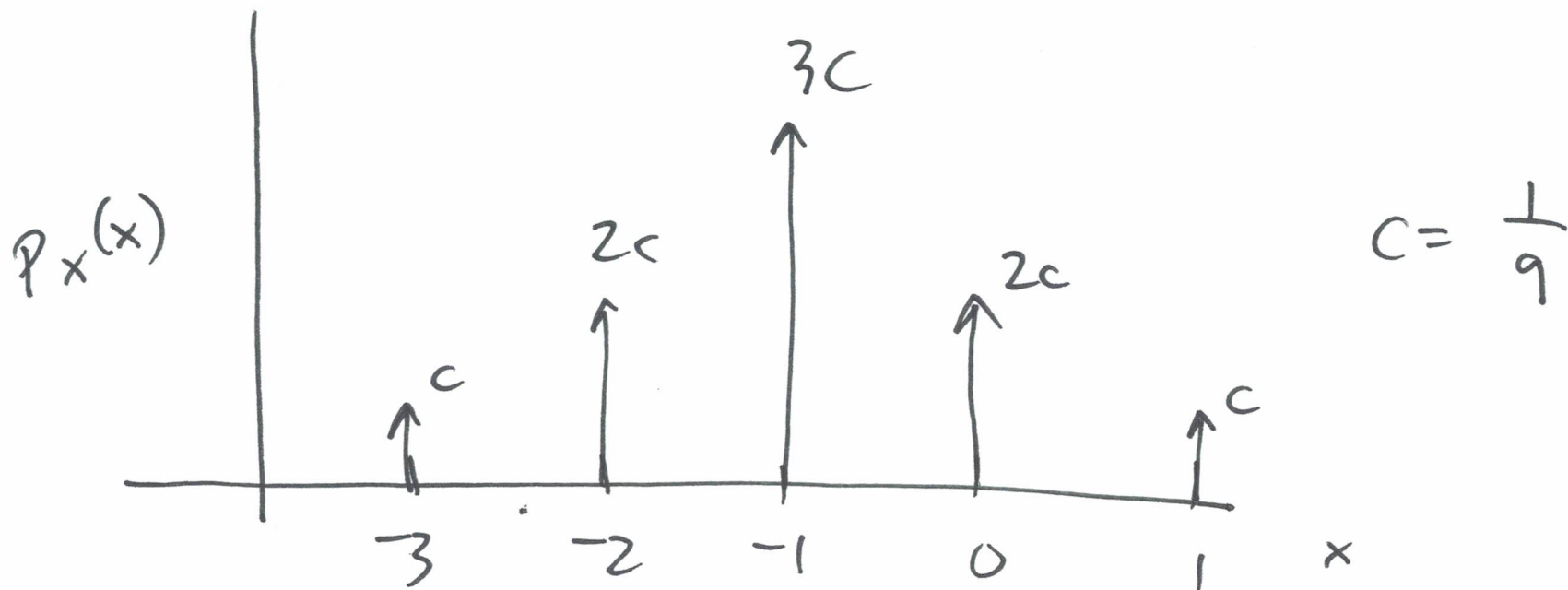
$E(X^m)$ IS CALLED THE
 m^{th} MOMENT OF X .

$E((X - \mu_x)^m)$ IS THE m^{th}
CENTRAL MOMENT OF X .

THE PMF OF A RV IS GIVEN BY:

$$P_X(n) = \begin{cases} c(3 - |n+1|) & n \in \{-3, -2, -1, 0, 1\} \\ 0 & \text{OTHERWISE.} \end{cases}$$

a) WHAT VALUE OF c MAKES THIS A VALID PMF?



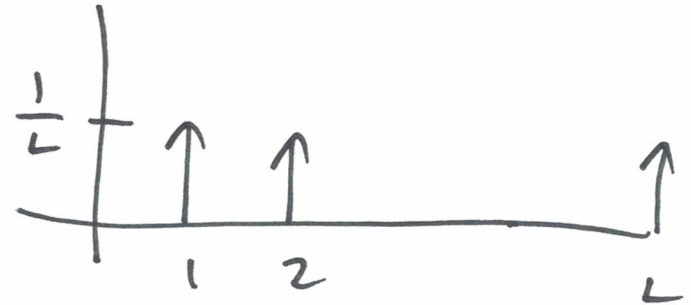
$$E(X) = -3\left(\frac{1}{9}\right) + -2\left(\frac{2}{9}\right) + -1\left(\frac{3}{9}\right) + 0 \cdot \frac{2}{9} + 1 \cdot \frac{1}{9}$$

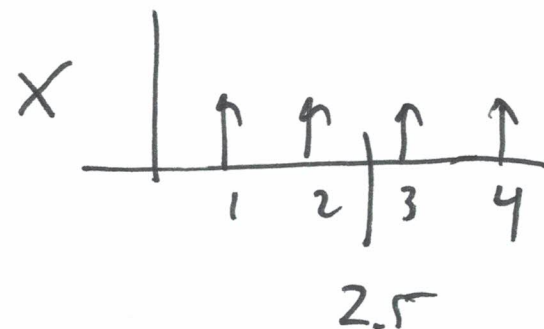
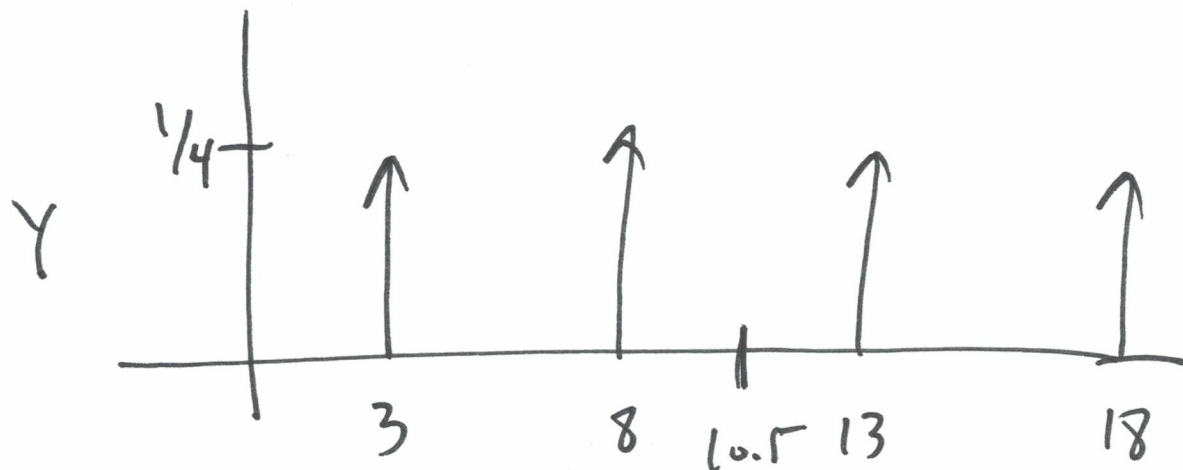
$$= -1.$$

USEFUL: VARIANCE OF A UNIFORM R.V.
ON $\{1, 2, \dots, L\}$

$$E(X) = \frac{L+1}{2}$$

$$\text{Var}(X) = \frac{L^2 - 1}{12}$$





IF X IS UNIFORM ON $\{1, 4\}$

$$Y = 5X - 2$$

$$E(Y) = 5E(X) - 2 = 5\left(\frac{5}{2}\right) - 2 = 10.5$$

$$\begin{aligned} \text{Var}(Y) &= \text{Var}(5X - 2) \\ &= \text{Var}(5X) = 5^2 \text{Var}(X) = 25 \cdot \frac{4^2 - 1}{12} \end{aligned}$$